

CS 3563: Introduction to DBMS II

Homework 3-4: Query Processing and Optimization/Database Design

Due Date and Time: 9 March. 2026, 11:59 PM IST

Instruction

Total Marks: 45

1. The homework must be completed individually.
2. Referring to external resources beyond the class notes and reference text-books/papers is not permitted.
3. The submission must be a single PDF and submitted via Moodle.
4. Late submissions will be handled in accordance with the course policy.

Question 1

(10 + 3 = 13 Marks)

- (a) Let relations $R(A,B,C)$ and $S(C,D,E)$ have the following properties:
- R has 20000 tuples,
 - S has 45000 tuples,
 - 25 tuples of R fit into one block,
 - 30 tuples of S fit into one block.

Estimate the number of block accesses required, using each of the following join strategies for equi-joining R with S on attribute C , which is uniformly distributed over a large domain. (For the first four strategies, assume that 3 buffers are available from the system).

- (i) Tuple-based Nested Loop (assume tuples for both the tables are spread across blocks and need not be stored contiguously)
 - (ii) Block Nested Loop
 - (iii) Sort-Merge Join (assume tables are unsorted initially)
 - (iv) Index Join (assume B^* -tree index on join attribute C exists only for relation R and that up to twenty pointers fit in an index page)
 - (v) Hash Join (assume that a pair of hash partitions can simultaneously fit in memory)
- (b) Ordinary joins can be processed using nested loops, sort-merge, and hash-join algorithms. Which of these algorithms can be used to compute the left outer join? Explain.

Question 2

(8 + 12 = 20 Marks)

Consider a System-R database platform hosting the EMP (Name, Age, Salary, CSY, RSY) table, which is physically sorted on the Salary attribute. Here, CSY and RSY stand for Completed Service Years and Remaining Service Years, respectively. The following query is issued by a user:

```
select * from EMP where Salary > 40000 and Age = 45 and CSY = RSY
```

Assume the following:

- B+ tree indexes I1 on Salary, I2 on Age, and I3 on RSY are available.
- Salary ranges from 20000 to 100000, Age ranges from 21 to 60, and RSY ranges from 5 to 29.
- The buffer pool allocated to the query is of size 1200 pages.
- The statistics on EMP are as follows (refer the paper shared on Moodle for term definitions):
 - Data Related: NCARD = 100000 tuples, TCARD = 1000 pages, P = 0.8,
 - Index I1 related: ICARD(I1) = 1000 values, NINDX(I1) = 500 pages,
 - Index I2 related: ICARD(I2) = 40 values, NINDX(I2) = 20 pages,
 - Index I3 related: ICARD(I3) = 25 values, NINDX(I3) = 10 pages
- Only IOs (i.e. disk accesses) are taken into account in the cost computations.

For the above framework,

- (a) Calculate the tuple cardinality estimated by System R for the query's result.
- (b) What access paths would be considered by the optimizer for the relation EMP? Which path is chosen? Show the cost calculations.

Question 3

(12 Marks)

For a relation scheme $R = \{ACDGLPRSTX\}$, the following functional dependencies hold:

$$\begin{aligned} S &\rightarrow ALX & C &\rightarrow P & DRT &\rightarrow C \\ DPT &\rightarrow R & DST &\rightarrow R & CS &\rightarrow G \\ R &\rightarrow T & L &\rightarrow A & GR &\rightarrow DS \end{aligned}$$

Your assignment partner claims that the following set of relations is a lossless-join, BCNF, dependency-preserving decomposition of R:

$$CP \quad RT \quad AL \quad CDR \quad LSX \quad DGRS$$

Do you agree? Explain.